

Q1. The given project Q1 already contains statements to implement a simple program using Binary Search Tree (BST) structure. In this exercise, the BST only contains the nodes with $key < 100$. You should write statements to the following functions:

a. **insert**(*k*) and **f1**(*k*) :

- Insert a new node with **key=k** to the BST.
- Search number **k** in the Binary Search Tree. Return **True** if this number exists in the BST, and **False** if it does not exist.

Expected result: True

b. **f2**() : The height of a tree is the number of edges on the longest path from the root to a leaf node. Calculate the height of the Binary Search Tree.

Expected result: 3

c. **f3**() : Count the amount of leaf nodes in the Binary Search Tree with **key>10**.

Expected result: 2

d. **f4**() : Compute the sum of all leaf nodes of Binary Search Tree.

Expected result: 70

e. **f5**(*k*) : Insert an even number to the Binary Search Tree, then return the height of the new BST. Hint: (1) you should check whether the number **k** can be inserted to the BST or not, (2) then you decide to insert **k** to the BST or not, (3) finally, calculate the height of the BST.

Expected result: 4

Notes:

- Do not edit given statements in the Main class.
- You can create new functions if you see they are necessary.